Ritwick Verma

Search & Retrieval Infrastructure | C++ | Systems
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SUMMARY

Search infrastructure engineer with 5.5 years on Cassini, eBay's C++ search engine serving every ebay.com query across 1B+ listings. Owns fitment end to end and builds retrieval, caching and query-transform features shipped through A/B experiments; 215 merged PRs, top ~10% of 121 contributors. Builds side projects by directing AI coding agents, and verifies every result.

EXPERIENCE

eBay

Software Engineer 3 (promoted), Search Backend (Cassini) — San Francisco, CA — Jun 2023 – Present

- **Own eBay Motors fitment**, the C++ subsystem that decides which parts fit a buyer's vehicle, across 16 sites and 19 catalogs; top author of the codebase (31% of lines, 1.6x the next contributor). Designed the query operator clients use to request fit constraints or per-item "fits" signals.
- **Retrieval and caching:** built a request-scoped backend result cache that removes redundant round trips across recall-recovery passes; client-supplied embeddings that skip the embedding service; request batching; restored an embedding cache that had silently stopped hitting.
- **Designed 3 C++ query transforms** for product-grouped results-page carousels, validated by production-traffic replay; fused 3 ranking expressions into one native macro, proven identical to the deployed artifact.
- **Made the tokenizer stack allocator-aware (PMR)** in a 7-PR refactor, then shipped it as an HTTP API that other teams call.
- **Reliability:** caught an init-order bug before release by replaying 50,000 production-sampled queries (new errors down ~96%); hotfixed a production parser crash; capped a combinatorial blow-up in query rewriting.
- **Developer productivity:** wrote the markdown-driven test frameworks the team adopted as standard, a language server for Cassini's query language (719 operators) and a multi-environment query CLI.

Software Engineer 2, Search Backend (Cassini) — San Jose, CA — Mar 2021 – Jun 2023

- **Rewrote fitment for garage vehicles and tire/wheel specifications** (88 merged PRs in 2022), with cross-site spec translation for DE/FR/IT; skipped downstream service calls when the vehicle is known; added A/B experimentation to fitment.
- **Made the engine's core query, table and property containers allocator-enabled** (C++ polymorphic allocators), groundwork used across the codebase.

Earlier: Software intern at Shoelace Wireless (Android, 2020) and BHEL (Java EE, 2017).

SELECTED PROJECTS

Projects built by directing AI coding agents (Claude Code and my own agent); I set the design and verify the results. Also: ShutUp, an Android app with 26,000+ downloads (v1 hand-written, 2016).

- **Self-hosted agent stack:** An AI agent on my homelab with tools for code, mail, documents and infrastructure; it builds, operates and monitors everything below while I set direction, review diffs and verify results.
- **TizenAmbilight:** In-TV ambilight for Samsung Tizen: the public capture API returns only the UI layer, so it reads the undocumented video plane instead; PQ-HDR tone mapping and BT.2020 gamut mapping at 20–25 fps, decoders verified byte-for-byte against libjpeg-turbo.
- **Crush 80 firmware:** Open ZMK firmware for a keyboard whose vendor never released source: Telink RISC-V SoC, 154 per-key LEDs on a ported SPI-DMA driver, pinout recovered from PCB photos and the stock binary.
- **Homelab as code:** 17 Proxmox guests behind Traefik, Authelia SSO and a Headscale VPN; 15 declared in OpenTofu, nightly CI that commits config drift, alerts once per real fault.
- **Plutus:** Quant research harness designed to falsify itself: point-in-time backtests, block-bootstrap confidence intervals, five popular ideas tested and rejected with evidence.

SKILLS

- **Languages:** C++17/20/23, Python, Java, Kotlin, C#, C
- **C++ tooling:** Polymorphic allocators (PMR), ASan, clang-tidy, CMake, GoogleTest
- **Search & IR:** Query rewriting, tokenization, recall and ranking factors, embedding retrieval, caching, A/B experimentation
- **AI engineering:** Directing coding agents (Claude Code), agent tooling and MCP servers, review and verification workflows
- **Systems:** Linux, Docker, Proxmox, OpenTofu, GitLab CI, Traefik/OIDC

EDUCATION

M.C.S., Computer Science — University of California, Irvine — 2019 – 2020
B.Tech, Computer Science & Engineering — SRM Institute of Science and Technology — 2015 – 2019